SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2– Concept Mapping (5 QUESTIONS)

1.Construct a comprehensive and well-organized concept map illustrating the complete pipeline from real-world scene capture to digital image representation. Identify all major concepts, establish meaningful relationships among intermediate transformations and dependencies, integrate insights from at least five recent literature sources, and present the map with clear hierarchy, linking phrases, and logical organization.

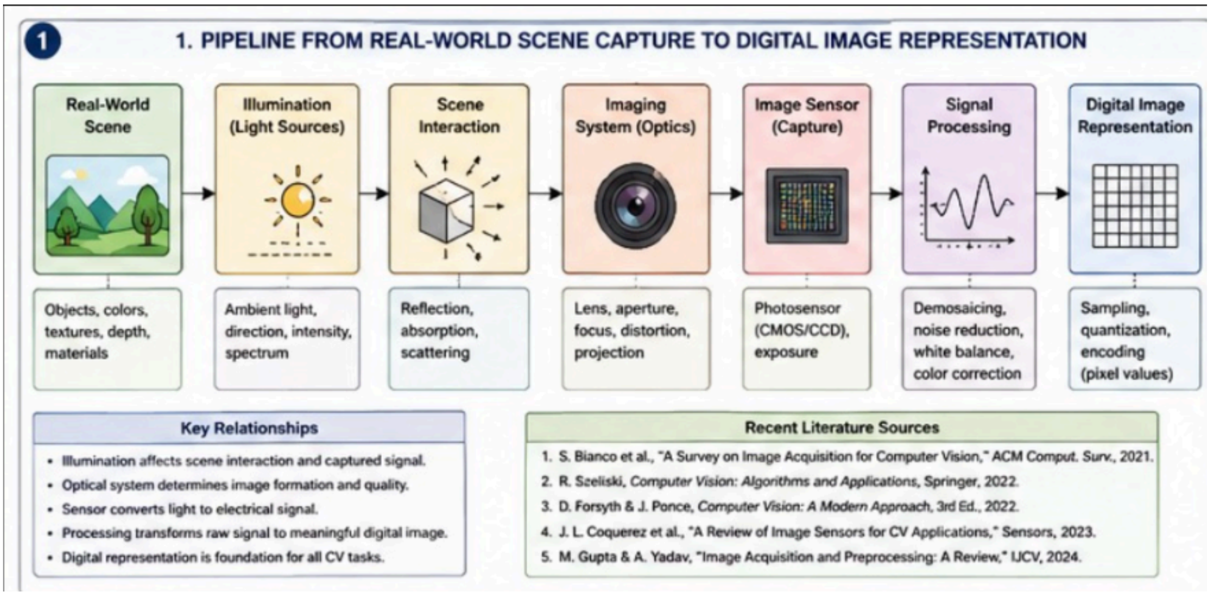
2.Develop a hierarchically structured concept map illustrating the major stages of visual data handling in a Computer Vision system. Clearly show the relationships among acquisition, preprocessing, feature extraction, analysis, and interpretation stages, explain their contributions to meaningful outputs, incorporate findings from at least five literature sources, and ensure clarity and readability.

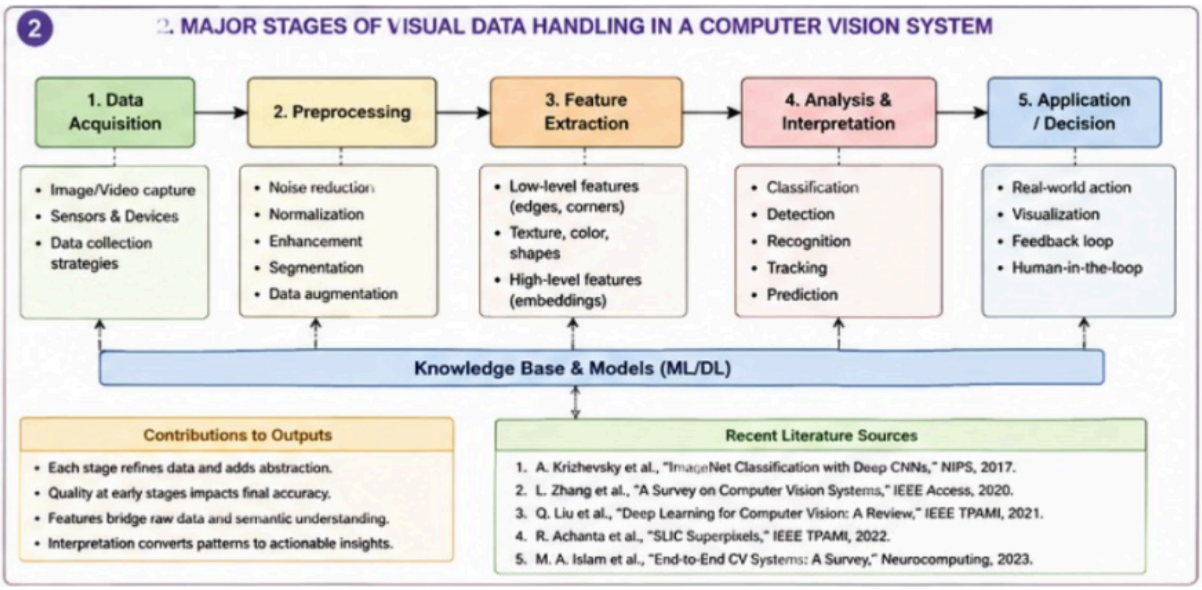
3.Create a detailed concept map explaining the relationships among image resolution, spatial resolution, intensity resolution, and image quality. Clearly illustrate cause-and-effect relationships, discuss their impact on image usability in different applications, integrate evidence from at least five research articles, and organize the concepts logically.

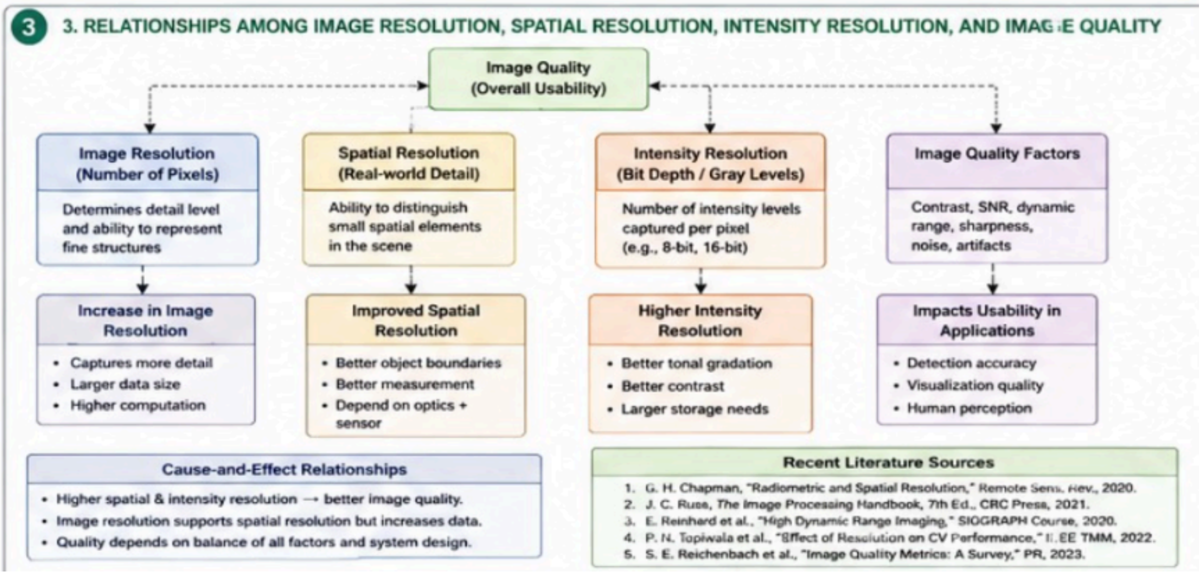
4.Design an analytically organized concept map illustrating the factors influencing image acquisition conditions, including illumination, sensor characteristics, motion, environment, and optics. Show how these factors affect downstream image processing and application performance, integrate relevant literature findings, and present the map in a visually coherent manner.

5.Construct an original concept map illustrating the relationship between Computer Vision system capabilities and the selection of appropriate vision-based applications. Identify major system capabilities, establish their influence on application effectiveness, integrate perspectives from multiple literature sources, and present the concepts using a clear hierarchical structure.

Assessment Tool -2



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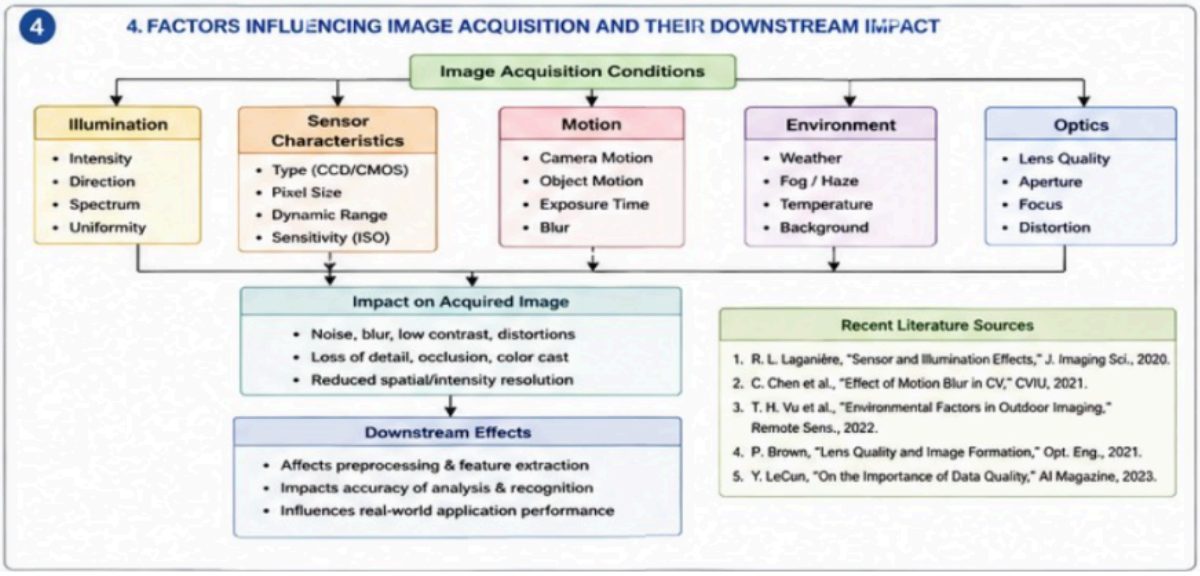
3. RELATIONSHIPS AMONG IMAGE RESOLUTION, SPATIAL RESOLUTION, INTENSITY RESOLUTION, AND IMAGE QUALITY
Image Quality (Overall Usability)
Image Resolution (Number of Pixels)
Determines detail level and ability to represent fine structures
Spatial Resolution (Real-world Detail)
Ability to distinguish small spatial elements in the scene
Intensity Resolution (Bit Depth / Gray Levels)
Number of intensity levels captured per pixel (e.g., 8-bit, 16-bit)
Image Quality Factors
Contrast, SNR, dynamic range, sharpness, noise, artifacts
Increase in Image Resolution
• Captures more detail
• Larger data size
• Higher computation
Improved Spatial Resolution
• Better object boundaries
• Better measurement
• Depend on optics + sensor
Higher Intensity Resolution
• Better tonal gradation
• Better contrast
• Larger storage needs
Impacts Usability in Applications
• Detection accuracy
• Visualization quality
• Human perception
Cause-and-Effect Relationships
• Higher spatial & intensity resolution → better image quality.
• Image resolution supports spatial resolution but increases data.
• Quality depends on balance of all factors and system design.
Recent Literature Sources
1. G. H. Chapman, "Radiometric and Spatial Resolution," Remote Sens. Rev., 2020.
2. J. C. Russ, The Image Processing Handbook, 7th Ed., CRC Press, 2021.
3. E. Reinhard et al., "High Dynamic Range Imaging," SIGGRAPH Course, 2020.
4. P. N. Topiwala et al., "Effect of Resolution on CV Performance," IEEE TMM, 2022.
5. S. E. Reichenbach et al., "Image Quality Metrics: A Survey," PR, 2023.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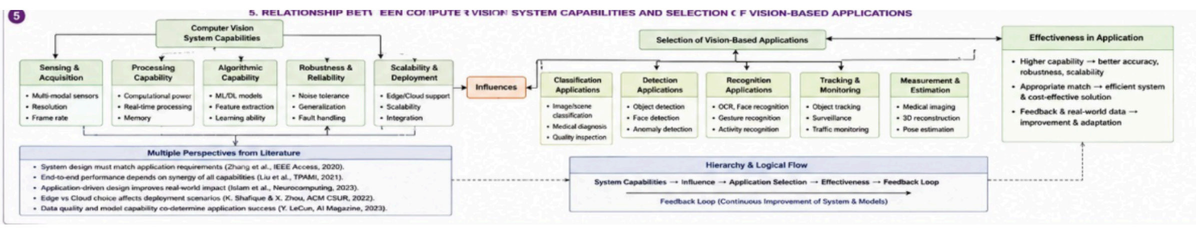
4. FACTORS INFLUENCING IMAGE ACQUISITION AND THEIR DOWNSTREAM IMPACT
Image Acquisition Conditions
Illumination
• Intensity
• Direction
• Spectrum
• Uniformity
Sensor Characteristics
• Type (CCD/CMOS)
• Pixel Size
• Dynamic Range
• Sensitivity (ISO)
Motion
• Camera Motion
• Object Motion
• Exposure Time
• Blur
Environment
• Weather
• Fog / Haze
• Temperature
• Background
Optics
• Lens Quality
• Aperture
• Focus
• Distortion
Impact on Acquired Image
• Noise, blur, low contrast, distortions
• Loss of detail, occlusion, color cast
• Reduced spatial/intensity resolution
Downstream Effects
• Affects preprocessing & feature extraction
• Impacts accuracy of analysis & recognition
• Influences real-world application performance
Recent Literature Sources
1. R. L. Laganière, "Sensor and Illumination Effects," J. Imaging Sci., 2020.
2. C. Chen et al., "Effect of Motion Blur in CV," CVIU, 2021.
3. T. H. Vu et al., "Environmental Factors in Outdoor Imaging," Remote Sens., 2022.
4. P. Brown, "Lens Quality and Image Formation," Opt. Eng., 2021.
5. Y. LeCun, "On the Importance of Data Quality," AI Magazine, 2023.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Computer Vision System Capabilities
Sensing & Acquisition
• Multi-modal sensors
• Resolution
• Frame rate
Processing Capability
• Computational power
• Real-time processing
• Memory
Algorithmic Capability
• ML/DL models
• Feature extraction
• Learning ability
Robustness & Reliability
• Noise tolerance
• Generalization
• Fault handling
Scalability & Deployment
• Edge/Cloud support
• Scalability
• Integration
Influences
Selection of Vision-Based Applications
Classification Applications
• Image/scene classification
• Medical diagnosis
• Quality inspection
Detection Applications
• Object detection
• Face detection
• Anomaly detection
Recognition Applications
• OCR, Face recognition
• Gesture recognition
• Activity recognition
Tracking & Monitoring
• Object tracking
• Surveillance
• Traffic monitoring
Measurement & Estimation
• Medical imaging
• 3D reconstruction
• Pose estimation
Effectiveness in Application
• Higher capability → better accuracy, robustness, scalability
• Appropriate match → efficient system & cost-effective solution
• Feedback & real-world data → improvement & adaptation
Multiple Perspectives from Literature
• System design must match application requirements (Zhang et al., IEEE Access, 2020).
• End-to-end performance depends on synergy of all capabilities (Liu et al., TPAMI, 2021).
• Application-driven design improves real-world impact (Islam et al., Neurocomputing, 2023).
• Edge vs Cloud choice affects deployment scenarios (K. Shafique & X. Zhou, ACM CSUR, 2022).
• Data quality and model capability co-determine application success (Y. LeCun, AI Magazine, 2023).
Hierarchy & Logical Flow
System Capabilities → Influence → Application Selection → Effectiveness → Feedback Loop
Feedback Loop (Continuous Improvement of System & Models)